EV Spawning Simulation — Findings

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1. Introduction

This document was made on the background that we wanted to investigate how we would be able to spawn EVs consistently through every interval / iteration of the simulation, but we wanted to avoid spawning more EVs than we had space allocated for.

These experiments tried to find a way of spawning the EVs in a way such a way that we don't go above the allocated space for EVs. For these simulations we attempted to keep the numbers of EVs below the total EVs in Denmark in January 2026, found by Statistics Denmark (DST)[1].

2. Simulation Design

2.1. CarsOnRoad

The CarsOnRoad class contains static information about how busy the roads are at a given point in a given day, together with a baseline of EVs (so there is not 0 EVs) and a maximum number allowed (the total number of EVs registered). To get how many EVs there should be on the road, the simulation looks up the congestion table of the index of the day and hour and performs a calculation for the number of EVs that should be on the road.

```
cars = cls.BASELINE_CARS +
  ( (cls.PEAK_CARS - cls.BASELINE_CARS) * car_congestion // cls._MAX_CONGESTION )
```

2.2. Spawn Fraction

At the time of writing this we decided that the simulation should spawn EVs every 30 minutes, so to accomodate for that a spawn fraction was implemented. Initially, this was just a multiplier of 0.5, such that if 100 EVs needed to be on the road in the interval 5-6, we spawn $100 * 0.5 = 50$ at 05:00, and then $100 * 0.5 = 50$ again at 05:30 to get the total 100 EVs between 5 and 6. The only difference that could impact results with this approach, is if the EVs journey completes before the next interval, meaning that there would be that many fewer EVs.

2.3. Event Loop

The EventLoop is the core of the simulation. It simulates one full week with 30-minute ticks. It is in charge of aging existing EVs (progressing their journey), spawning new EVs, and recording a snapshot of the system which stores the current active EV count, day, hour, and minute for later visualisation.

[1] Statistics Denmark (DST), January 2026. https://www.dst.dk/da/Statistik/udgivelser/NytHtml?cid=51885

3. Journey Duration Distribution

To see how the amount of short versus long journeys might affect the simulation, there was initially a distribution of short journeys that had a higher scalar than long journeys. This was also to appeal to realism, where there on average likely is far more journeys within a certain region of Denmark, than cross country.

```
MIN_JOURNEY_MINUTES = 10    # 10 minutes
SHORT_JOURNEY_MAX   = 120   # 2 hours
MAX_JOURNEY_MINUTES = 360   # 6 hours
SHORT_TRIP_WEIGHT   = 4.0
...
short_range = SHORT_JOURNEY_MAX   - MIN_JOURNEY_MINUTES   # 110 min
long_range  = MAX_JOURNEY_MINUTES - SHORT_JOURNEY_MAX      # 240 min
...
w_short = SHORT_TRIP_WEIGHT * short_range
w_long  = 1.0               * long_range
p_short = w_short / (w_short + w_long)
...
u = random.random()
if u < p_short:
    return int(MIN_JOURNEY_MINUTES + (u / p_short) * short_range)
else:
    return int(SHORT_JOURNEY_MAX + ((u - p_short) / (1.0 - p_short)) * long_range)
```

The range of the trips are multiplied by some weight factor, which for short journeys is 4.0 and for long journeys is none (or just 1.0). The probability of the short trip is then calculated in this line: `p_short = w_short / (w_short + w_long)` where the calculation would look like:

$$\text{w_short} = 4.0 * 110 = 440$$

$$\text{w_long} = 1.0 * 240 = 240$$

$$\text{p_short} = \frac{440}{680} \approx 0.647 \approx 65\%$$

4. Results

When running the simulation with a spawn fraction of 0.5 and the roughly 65% weight on short journeys, we received these results.

```
…/P8/CarsOnRoadPython ❯ python main.py
Starting simulation: 336 ticks, spawn_fraction=0.5
  Day 1/7 (   Sunday) — tick   0 — active cars:  14,344
  Day 2/7 (   Monday) — tick  48 — active cars: 132,790
  Day 3/7 (  Tuesday) — tick  96 — active cars: 161,168
  Day 4/7 (Wednesday) — tick 144 — active cars: 141,012
  Day 5/7 ( Thursday) — tick 192 — active cars: 141,158
  Day 6/7 (   Friday) — tick 240 — active cars: 147,234
  Day 7/7 ( Saturday) — tick 288 — active cars: 135,463

--- Simulation Summary ---
  Peak active cars :    870,455  (Tuesday 08:30)
  Exceeds TOTAL_EVS:    314,455
  Average active   :    324,354
  Shortest journey :         10 min
  Longest journey  :        359 min
  Average journey  :      125.6 min
--------------------------
```

Figure 1: Terminal output for first iteration. `TOTAL_EVS` is how many EVs past the allowed maximum amount of EVs.

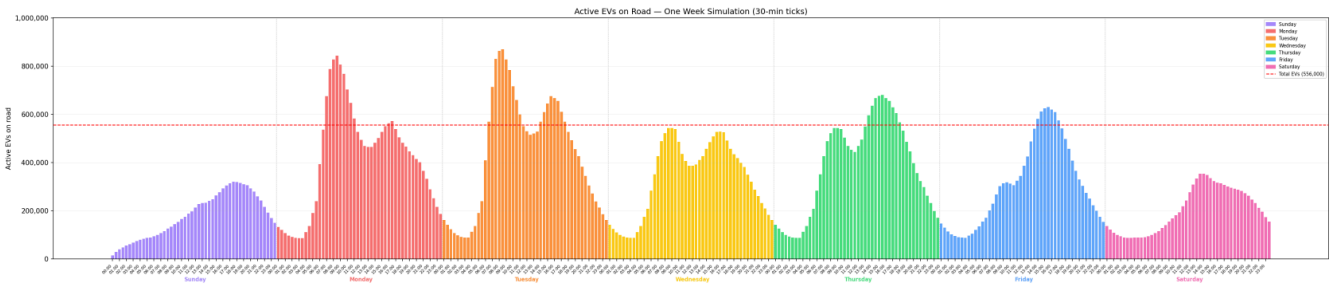

Figure 2: First iteration: Bar graph of amount of EVs on the road for each tick in the week. As can seen, the graphs on Monday, Tuesday, Thursday and Friday exceed the allowed limit.

Based on the results, the amount of EVs on the road quite well follows the expected curve of amount of EVs on the road at some given time, with many more EVs at busy hours, and relatively few EVs at quiet hours. With that said, it was obvious that the graphs exceeded the limit of EVs, and that by quite a lot.

To try and fix this, the spawn fraction scalar was lowered to **0.4**.

```
● …/P8/CarsOnRoadPython × python main.py
Starting simulation: 336 ticks, spawn_fraction=0.4
  Day 1/7 (   Sunday) — tick   0 — active cars:  11,475
  Day 2/7 (   Monday) — tick  48 — active cars: 106,598
  Day 3/7 (  Tuesday) — tick  96 — active cars: 128,941
  Day 4/7 (Wednesday) — tick 144 — active cars: 113,226
  Day 5/7 ( Thursday) — tick 192 — active cars: 113,124
  Day 6/7 (   Friday) — tick 240 — active cars: 117,847
  Day 7/7 ( Saturday) — tick 288 — active cars: 108,359

--- Simulation Summary ---
  Peak active cars :    696,328  (Tuesday 08:30)
  Exceeds TOTAL_EVS:    140,328
  Average active   :    259,502
  Shortest journey :         10 min
  Longest journey  :        359 min
  Average journey  :      125.6 min
--------------------------
```

Figure 3: Terminal output for second iteration. `TOTAL_EVS` is how many EVs past the allowed maximum amount of EVs.

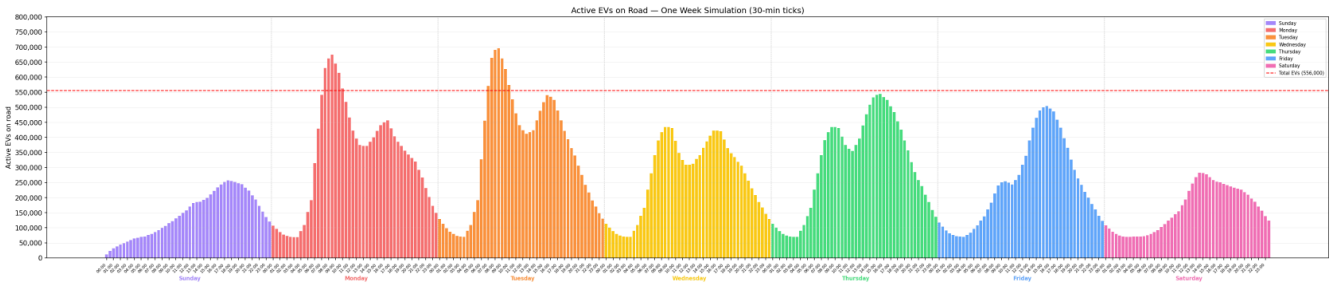

Figure 4: Second iteration: Bar graph of amount of EVs on the road for each tick in the week. As can seen, the bars on Monday, Tuesday still exceed the allowed limit.

These results showed that the amount of EVs that exceeded the limit lowered substantially, all the way down to $(140328 \div 314455) \times 100 \approx 44.6\%$, meaning we managed to more than half the exceeded amount.

But these results still exceed the allowed amount, so we lowered the spawn fraction to **0.3**.

```
…/P8/CarsOnRoadPython × python main.py
Starting simulation: 336 ticks, spawn_fraction=0.3
Total EVs=556,000, short trip weight=4.0x
Trip probability: 65% short (10-120 min) vs 35% long (120-360 min)
  Day 1/7 (   Sunday) — tick   0 — active cars:   8,606
  Day 2/7 (   Monday) — tick  48 — active cars:  79,619
  Day 3/7 (  Tuesday) — tick  96 — active cars:  96,840
  Day 4/7 (Wednesday) — tick 144 — active cars:  84,407
  Day 5/7 ( Thursday) — tick 192 — active cars:  84,687
  Day 6/7 (   Friday) — tick 240 — active cars:  88,448
  Day 7/7 ( Saturday) — tick 288 — active cars:  81,032

--- Simulation Summary ---
  Peak active cars :    521,930  (Tuesday 08:30)
  Exceeds TOTAL_EVS:    -34,070
  Average active   :    194,664
  Shortest journey :         10 min
  Longest journey  :        359 min
  Average journey  :      125.7 min
  Total journeys   : 11,474,315
  Amount of short trips spawned (<= 120 min):  7,470,452
  Amount of long trips spawned (> 120 min):  4,003,863
--------------------------
```

Figure 5: Terminal output for third iteration. `TOTAL_EVS` is now a negative number, which means we did not exceed the allowed limit. Additionally, the amount of total journeys, the amount of short journeys, and the amount of long journeys were added to the logs.

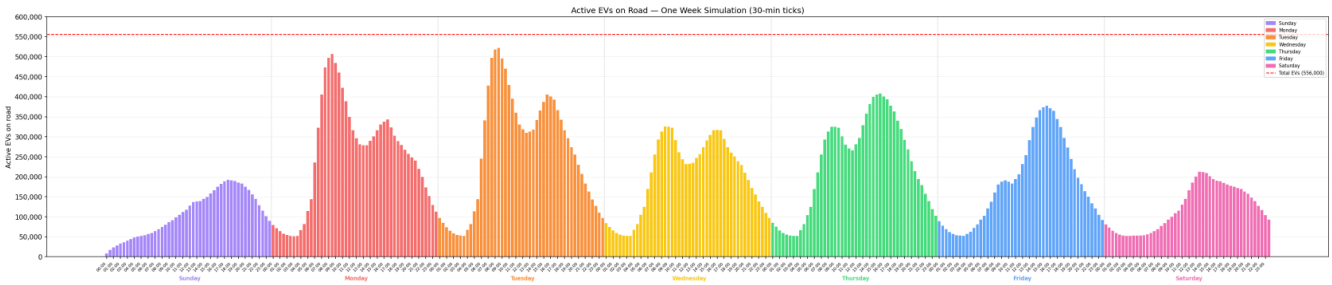

Figure 6: Third Iteration: Bar graph of amount of EVs on the road for each tick in the week. As can seen, the graphs did not exceed the allowed limit, as expected from the terminal output.

These settings resulted in the limit not being exceeded. Here the amount of long and short journeys were logged, and perhaps that weighting might have affected whether this is the correct scalar, as longer journeys would take longer to finish, and therefore stay on the roads for longer. So for the fourth iteration, the weight was lowered to 3.0, which gives: w_short $= 3.0 * 110 = 330$

w_long $= 1.0 * 240 = 240$

p_short $= \frac{330}{570} \approx 0.579 \approx 58\%$

```
● …/P8/CarsOnRoadPython ❯ python main.py
  Starting simulation: 336 ticks, spawn_fraction=0.3
  Total EVs=556,000, short trip weight=3.0x
  Trip probability: 58% short (10-120 min) vs 42% long (120-360 min)
    Day 1/7 (   Sunday) — tick   0 — active cars:   8,606
    Day 2/7 (   Monday) — tick  48 — active cars:  88,127
    Day 3/7 (  Tuesday) — tick  96 — active cars: 108,608
    Day 4/7 (Wednesday) — tick 144 — active cars:  93,670
    Day 5/7 ( Thursday) — tick 192 — active cars:  93,909
    Day 6/7 (   Friday) — tick 240 — active cars:  98,160
    Day 7/7 ( Saturday) — tick 288 — active cars:  89,385

  --- Simulation Summary ---
    Peak active cars :    549,520  (Tuesday 08:30)
    Exceeds TOTAL_EVS:     -6,480
    Average active   :    208,298
    Shortest journey :         10 min
    Longest journey  :        359 min
    Average journey  :      137.5 min
    Total journeys   : 11,465,316
    Amount of short trips spawned (<= 120 min):  6,686,444
    Amount of long trips spawned (> 120 min):  4,778,872
  --------------------------
```

Figure 7: The terminal for the fourth iteration, with the short-journey weight lowered from 4.0 to 3.0.

With this change the limit was still not exceeded, but still raised by roughly 10.000 EVs. Removing the weight for short journeys completely might result in exceeding the allowed limit.

```
…/P8/CarsOnRoadPython ❯ python main.py
Starting simulation: 336 ticks, spawn_fraction=0.3, Total EVs=556,000
  Day 1/7 (   Sunday) — tick   0 — active cars:   8,606
  Day 2/7 (   Monday) — tick  48 — active cars:  87,957
  Day 3/7 (  Tuesday) — tick  96 — active cars: 108,378
  Day 4/7 (Wednesday) — tick 144 — active cars:  93,672
  Day 5/7 ( Thursday) — tick 192 — active cars:  93,527
  Day 6/7 (   Friday) — tick 240 — active cars:  98,251
  Day 7/7 ( Saturday) — tick 288 — active cars:  89,541

--- Simulation Summary ---
  Peak active cars :    549,165  (Tuesday 08:30)
  Exceeds TOTAL_EVS:     -6,835
  Average active   :    208,218
  Shortest journey :         10 min
  Longest journey  :        359 min
  Average journey  :      137.5 min
  Total journeys   : 11,465,376
--------------------------
```

Figure 8: The terminal for the fifth iteration, with weight for short journeys completely removed.

But actually this was not the case. Instead, the result ended up being that there were fewer EVs on the road total. But this could be a result of randomness playing into the favor of the experiment. To eliminate that, we ran the simulation 200 times, to see if there would ever be a run that would exceed the limit.

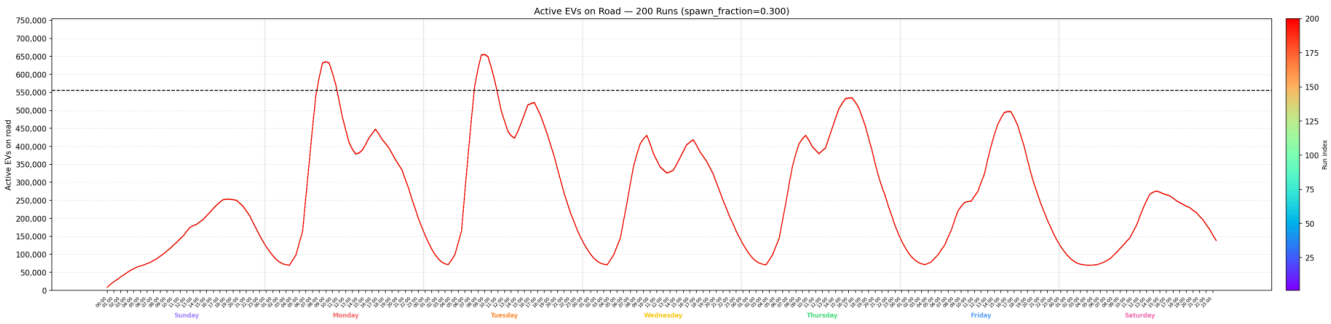

Figure 9: Multi-line graph of amount of EVs on the road for 200 simulations with a 0.3 spawn fraction and no weights on length of journeys.

The results here ended up showing that the limit was exceeded every time on Monday and Tuesday. What this rather large change came from is still not known. Additionally, we'd like to point out that the seemingly lack of variance is due to the law of large numbers, and the scale of the graph. The variance between the runs are not large enough to be seen in the scale of hundreds of thousandths, even if that variance is still some many thousandths.

But this evidently meant the spawn fraction had to be lowered. This iteration we wanted to lower it down to 0.25 from 0.3.

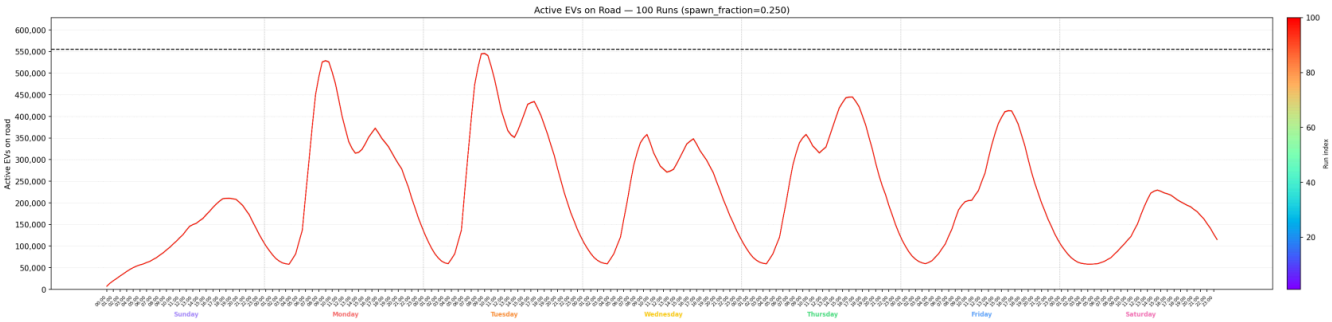

Figure 10: Multi-line graph of amount of EVs on the road for 200 simulations with a 0.25 spawn fraction.

And as can be seen by the results, none of the simulations ever exceed the limit for total EVs. And as mentioned before, for the scale of this simulation, the law of large numbers will ensure that the limit will never be exceeded if we run the simulation with these settings. However, to force variance into it, we ran the simulation another 200 times, with each simulation having each EV have a 50% chance when spawning to follow a mean journey time. This journey time was also randomized, and could be anything between 10-360.

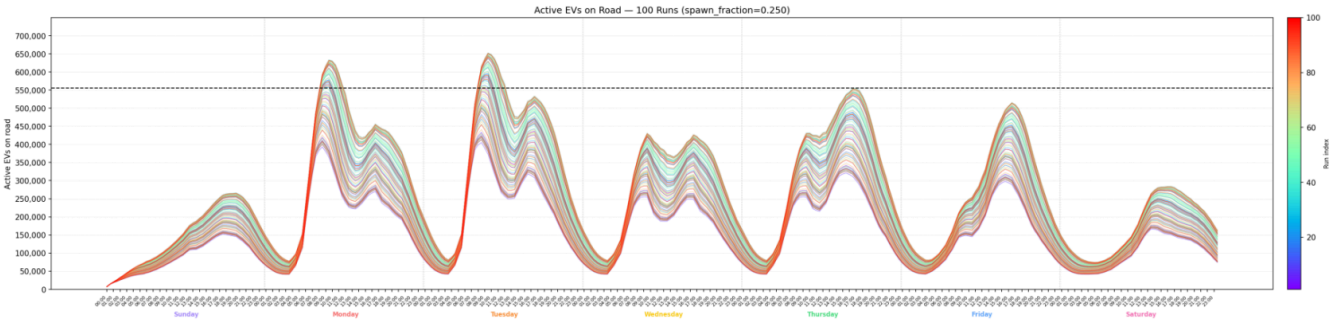

Figure 11: Multi-line graph of amount of EVs on the road for 200 simulations with a 0.25 spawn fraction and a 50% chance to spawn with a randomly allocated mean journey time between 10 and 360 minutes.

The results showed that there could be some runs where the allowed limit is exceeded. However, the odds of this ever happening is astronomically low, and even then, the chart shows that if we allocate somewhere slightly above 650.000 entries for our 556.000 EVs we should even in theory never have more EVs on the road than allocated space for them.

5. Conclusions

Based on the experiments made we found that spawning a fraction of EVs of however many EVs should be on the road at a given interval will result in the EVs not exceeding the total allocated space for our EVs, even if that allocated space is only exactly how many EVs exist in Denmark, even if journeys are not weighted such that there are more shorter journeys, e.g from home to the grocery store, than there are long journeys, e.g from Aalborg to Copenhagen.

We found that the spawn fraction should be 0.25, with a 30 minute interval of spawning.